Making a Habit of Habitat

There might have been some bumps along the way, but a collaboration between a nonprofit, a builders association, and a major building supply company yielded some efficient, comfortable, and affordable results.

by Daran Wastchak

Habitat for Humanity has successfully built affordable homes across several continents, largely because it uses donated materials and labor. In Phoenix, Arizona, those donations are helping Habitat to make homes that are also highly energy efficient. Operating under a tight budget and with volunteers behind the hammers, Habitat is building Villas Esperanza—one of the largest single communities of Energy Star homes that the nonprofit has ever built. In the words of Scott Balk, former construction manager for the Phoenix area Habitat affiliate Habitat Valley of the Sun, "Energy Star was more than a good idea. Although the extra costs were not insignificant, the good will demonstrated by Habitat and return for the family in particular, was just too good to pass up."

Improving Quality and Energy Efficiency

Planning for Villas Esperanza may have begun in 2001, but its origins can be traced back to 1996. That's when Home Depot in Phoenix sponsored its first Habitat home at the South Ranch community in south Phoenix. Although this and subsequent homes built through Home Depot sponsorship were built to Habitat's standard design, Home Depot challenged itself to do something more innovative with the home it opted to build in 2000 (see Table 1, p. 23). For assistance, it first turned for help to the Arizona Department of Commerce Energy Office.

A particular challenge was the duct system, which was shared by the evaporative cooler and the heat pump. Testing by the Energy Office on one Habitat home showed total system leakage of 440 CFM_{25}, with most of the leakage occurring at the barometric damper. The Energy Office noted in this instance that after only two years the barometric damper had so much water corrosion (primarily hard water buildup) that it had stopped closing properly. The logical recommendation was to remove the evaporative cooler, saving money by lowering utility bills as a result of reduced leakage and eliminating the up-front costs of the cooler (see "A/C Versus Evaporative Cooling," p. 22). Unfortunately, removing the cooler was not possible because the development's code restrictions required that each home have an evaporative cooler. Some home buyers insisted on having the cooler because their neighbor's house had one and they felt that evaporative coolers were an important component of the home.


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(top) A family gets set to enter their new, energy-efficient home. *(bottom)* A new Habitat community in Phoenix features all Energy Star homes.

Since it was unable to eliminate the evaporative cooler, the Energy Office recommended that a separate duct system be installed for the evaporative cooler on the Habitat-sponsored home. Because Home Depot was willing to cover the added cost for the separate duct system (approximately 30% over a single system), Habitat agreed to allow the change. Additional improvements to the home included jump ducts for room pressure relief. Jump ducts are short ducts, typically running from the ceiling in a bedroom up into the attic and back to the ceiling in a hallway, that provide an alternative pathway for supply air to exit a room when the door is closed. Other improvements included loose-fill fiberglass insulation in the walls and low solar gain low-e glass in the windows. Final testing showed that duct leakage for the A/C side of the system had been reduced to 32 CFM_{25} for the 3.5-ton system, well below the target of 70 CFM_{25}.

Introducing Energy Star

With so many quality and energy efficiency improvements to the home, Home Depot and the Energy Office called me to complete a plan analysis so that the home could be labeled as an Energy Star home. According to the EPA, the Home Depot Habitat home built in 2000 was one of the first Energy Star-labeled homes built by a nonprofit in the state of Arizona. The final HERS score for the home was 88, higher than the Energy Star minimum of 86.

According to Dow Riggler, community affairs representative for Home Depot, the company's interest in the Energy Star Homes program stemmed from its own energy efficiency program, called E+, through which they were already selling and promoting Energy Star and other energy-efficient products. The improvements necessary to achieve the Energy Star label helped Home

Depot to differentiate their home from the other 35 homes that Habitat Valley of the Sun builds each year. "It's nice to build affordable housing," Riggler says, "but why not also make these homes truly energy efficient? It only makes sense." Home Depot also viewed its participation in the Energy Star Homes program as a platform from which to educate the public on the importance of energy-efficient, quality construction.

Following the success of the 2000 home, Home Depot built a second and final Energy Star-labeled home at the South Ranch community. This home was completed in early 2001. The only changes from the Habitat standard design that were incorporated into the remaining 20 homes at South Ranch were air sealing and low-e glass in the windows. Upgrading to low-e glass for windows in these homes was made possible by a special price break from Home Depot.

Looking At Energy Star as a Standard

Planning for Habitat Valley of the Sun's newest community, Villas Esperanza, began in 1996. Villas Esperanza is being developed in south Phoenix, on land donated by Star Dust Development, and includes 95 lots. At the urging of Home Depot, Habitat made a commendable commitment in 2001 that all 95 of the homes at Villas Esperanza would be Energy Star labeled.

In order to achieve the Energy Star for Homes label for the entire community, Habitat is making low-e glass standard in all windows, and it continues to use SEER-12 heat pumps. Habitat has eliminated the evaporative coolers (and any need for a separate duct system); is making certain that the house framing is built properly for insulation (by capping all drop soffits and chases); is installing all insulation with careful attention to compression, gaps, voids, and misalignment; and is having a minimum of one in seven homes tested by a third-party independent HERS rater. I donated the plan analysis and Home Depot is paying for the cost of testing and inspections, which I am also conducting.

As a full partner in the Energy Star Homes program, Habitat for Humanity is being recognized for its efforts by participating in the EPA's Arizona Cooperative Advertising campaign in 2003 (it also participated in 2002). It is listed on the regional Energy Star Web site, www.ArizonaEnergyStarHomes.com, along with the many other Energy Star Homes builders in Arizona.

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Jesse Rowley, the operations manager for D.R. Wastchak, helped to build the reverse soffit in the Home Depot-EEBA home, moving ducts into conditioned space.

Yet Higher Hurdles

Not content to rest on their laurels, Habitat and Home Depot teamed up with the Energy and Environmental Building Association (EEBA) to build an even better mousetrap (see Table 1). Beginning with its 2000 conference in Denver, EEBA decided that partnering with local Habitat affiliates would provide an opportunity to introduce new technologies to Habitat and at the same time showcase the technologies for the conference attendees. Since then, EEBA has been using the occasion of its annual conference to help Habitat for Humanity complete a house using free labor and technical assistance from EEBA's members and donations of products from conference sponsors. The first home built at Villas Esperanza—the Home Depot-EEBA home—would go beyond the Energy Star standards and incorporate building design elements necessary to achieve the U.S. Department of Energy's Building America status (see "Putting Technology into Practice," *HE* March/April '01, p. 42). Getting to Building America status made it necessary, first and foremost, to move all of the ductwork in the attic to within the conditioned envelope in order to reduce the heat load on the ducts in the summer and reduce the impact of air leakage from the ducts. This proved to be one of the greatest challenges for this project.

Redesign began in the summer of 2002 with the assistance of Building Science Corporation (BSC), a Building America Partner. With an improved building envelope, the low solar gain low-e windows, and the ducts moved into conditioned space, *Manual J* calculations on the selected floor plan indicated that the air conditioning system could be reduced in size from the originally designed 3 tons to 2.5 tons. The HERS score for the home as designed for Energy Star and Building America was 89.9.

The typical method in Phoenix for moving ductwork within the conditioned space is to cathedralize the attic insulation inside an unvented attic. To cathedralize attic insulation is to seal the attic and move the insulation from the ceiling to the roof deck, so that the ducts in the attic are now below rather than above the insulation. Unfortunately, cathedralizing the insulation was not possible in this instance because the roofs at Villas Esperanza are covered with asphalt shingles rather than clay tile. Research by BSC and others shows that asphalt shingle roofs with insulation placed on the underside of the plywood sheathing can grow hot enough to shorten the life of the shingles. However, clay tile roofs have been shown to be much cooler than asphalt shingle roofs, making them suitable for cathedralized insulation applications in hot-dry climate areas like Phoenix.

The next option was to move the ducts into drop soffits within the conditioned space, air seal the soffits at the ceiling line, and insulate over the top. A full soffit and duct design layout was engineered by BSC and was agreed to by all parties—except the future homeowners who did not want any of the ceiling heights lowered beyond what was specified in the plans.

They say that the third time is the charm, and so it was with our final and least palatable option—building a reverse soffit into the attic, air sealing it, and insulating the sides and top. The challenge of this option was the complexity of the soffit framing, which the Habitat staff was not familiar with, and which volunteer carpenters would probably not be able to build. It was not easy, but a team of volunteers from my company worked with the Habitat staff over the course of two days to build the soffit.

Many of those involved with the Home Depot-EEBA home agreed that it was a real success. This success was due, particularly, to the sponsorship of Home Depot and Masco Contractor Services, and significant material and labor donations from Andersen Windows, Gale Insulation, and Chas Roberts Air Conditioning. Participants at the EEBA conference had an opportunity to volunteer time working at the house during the week of the conference, and a tour of the house under construction was coordinated as a preconference event. The home was dedicated in November 2002. But although the Home Depot-EEBA home was a success, there are also a number of lessons to be learned from this project. These lessons are worth noting for the future.

After all the work that went into getting the Habitat home to the Building America standard, the staff at Habitat made it clear that many of the design changes made to the home would not be incorporated into other Habitat homes. Several of the volunteers from the EEBA team were very disappointed by this attitude. Although cost was an important factor, further investigation revealed that there were several reasons behind Habitat's decision.


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D.R. Wastchak donated all the testing and inspections for the Home Depot-EEBA home.

When asked about ways to improve the EEBA-Habitat partnership for the future, Dow Riggler stated that EEBA should have "done more homework" on the local Habitat organization. Had they done so, they would have learned that Valley of the Sun is one of the biggest and best Habitat affiliates in the nation. It prided itself on the fact that it had already made important changes to its house designs—changes that not only improved the homes, but reduced energy consumption enough to make it one of the first Habitat affiliates to build to the Energy Star standard. According to both Home Depot and Habitat, the EEBA redesign effort was very much appreciated, but lacked sensitivity.

Habitat felt that once it had agreed to work with EEBA, the changes necessary to get to the Building America standard were pushed so strongly that it had no choice but to go along with them. The clearest example of this was moving the ducts into conditioned space. It seemed very unlikely that Habitat could implement this design change in other homes, particularly because the change was so challenging to figure out, and ultimately to build.

Many of the problems with the EEBA redesign effort were logistical.

A/C Versus Evaporative Cooling

In 1995, the Arizona Department of Commerce Energy Office began to work with Habitat for Humanity in Phoenix, Arizona, to assist with energy audits and diagnostics to determine potential upgrades for Habitat's house designs. Major issues found were duct leakage and room pressure imbalances. Most of the duct leakage was through the barometric damper on the dual A/C evaporative cooling system, which shared the same ductwork.

A major obstacle to removing the evaporative cooler was the misconception that affordable housing must have evaporative cooling. To overcome this, the Energy Office presented data collected from the local electric utility, Arizona Public Service (APS), and the results of their own computer analysis (using CALRES, an hourly simulation software). The cost of a dual cooling system was compared with the cost of A/C alone; the comparison included the cost of water usage and duct leakage through the evaporative cooler's damper while the A/C system was running. The results showed negligible net savings when considering a 12-SEER heat pump (see Table A). Savings from evaporative cooling is further eroded in the summer during the rainy season, when humidity makes evaporative cooling ineffective and A/C is used instead. If one factors in the cost of maintaining the evaporative cooler and the cost of repairing damage to the roof from water runoff, the dual cooling system is not cost-effective.

Table A. A/C-Evaporative Cooler Savings over Heat Pump-Only Systems

Heat Pump—Cooler Combination	Annual Savings	Leakage Cost	Net Savings	Payback (Yrs)
8.5-SEER heat pump	$116	$53	$63	19
10-SEER heat pump	$ 77	$46	$31	39
12-SEER heat pump	$ 42	$38	$ 4	300

Notes: Annual Savings is the savings realized when changing from a heat pump only system to a heat pump and evaporative cooler combination system (APS study).
Cooler is aspen pad evaporative cooler (installed cost: $1,200).
Cooler water usage: 16.2 gal/h.
Cooler operating hours: cooler only = 2,005, with heat pump = 1,384.
Water cost per year: cooler only = $53.85, with heat pump = $37.13.
Leakage Cost assumes 280 CFM_{25} leakage at barometric damper.

Table 1. A Comparison of Design Features

Habitat For Humanity "Standard" Design Features: • Slab-on-grade • 2 x 4 wood frame walls • Asphalt shingle roof • Dual-pane clear-glass windows, aluminum frames • R-13 fiberglass batt insulation in walls • R-30 fiberglass blown insulation in attic • Ducts in unconditioned attic, R-4.2 insulation • Evaporative cooler on roof • 12-SEER heat pump packaged unit on roof	**South Ranch**			**Villas Esperanza**	
	"Standard" Design	**Home Depot-Energy Star Design**	**Improved Standard Design**	**Energy Star**	**Home Depot-EEBA Building America**
HVAC					
Tight duct system	no	yes	no	yes	yes
Separate ducts for evap and AC	no	yes	no	no	no
Ducts in conditioned space	no	no	no	no	yes
Evaporative cooler	yes	yes	yes	no	no
12-SEER heat pump	yes	yes	yes	yes	yes
Room pressure relief (jump ducts/transfer grilles)	no	yes	no	no	yes
Fresh-air system (at return)	no	no	no	no	yes
Insulation					
Framing ready for insulation (capped soffits, chases)	no	yes	no	yes	yes
Correctly installed insulation (no gaps, voids, misalignment)	no	yes	no	yes	yes
Fiberglass batt insulation in walls	yes	no	yes	yes	no
Blown insulation in walls (fiberglass or cellulose)	no	yes	no	no	yes
Envelope air sealing	no	yes	yes	yes	yes
Low-e glass	no	yes	yes	yes	yes
ENERGY STAR water heater	no	yes	no	yes	yes
Third-party testing and inspections during construction	no	yes	no	yes	yes
Number of Homes Built to Each Standard	**174**	**2**	**20**	**94**	**1**

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Because Villas Esperanza was a brand-new community, drawings were not available until relatively late in the redesign process. And getting the drawings approved by the city was such a long and drawn out procedure for Habitat that any recommended changes that required additional approval from the city (such as an unvented, cathedralized attic) caused Habitat to be very concerned. The pressure to meet the Building America standard became necessary only after sponsorship commitments had been made with the understanding that the house could be built to that level. It was not until late in the process that we discovered that getting to Building America would require such drastic measures.

As a member of the local host committee for the EEBA conference, and as the liaison between Habitat and the EEBA team, I can accept responsibility for some of EEBA's insensitivity. In hindsight, we should have done a better job of communicating directly with the Habitat board of directors about what we were hoping to accomplish. In addition, we should have better understood and accepted the scheduling uncertainties that Habitat was facing with completing their drawings and getting them approved by the city. Handling this issue differently could have made the experience more positive for both Habitat and EEBA. Lastly, even though EEBA had partnered with Habitat in other cities as part of its annual conference, this does not mean that EEBA should expect to be automatically accepted with open arms by every and any Habitat affiliate. That was what it had expected, obviously, in Phoenix.

Making a Habit of Energy Star

I am proud to be working with Habitat for Humanity in Phoenix. And I am very pleased that it has made the full commitment to build all of its homes to the Energy Star standard, demonstrating that energy-efficient, high-quality construction is obtainable in the affordable housing sector. We hope that other Habitat affiliates around the country, perhaps with the encouragement and assistance of the national organization, will follow Phoenix's example and strive for the Energy Star designation.

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